The Weight Between and the Exit Line

Cross-block bounds for the boundary census, and where opened matter leaves the disk

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Abstract

Cut a fermionic system into a region A and its complement B. The one-body correlation matrix C then has a cross block, and $X_A = C_{AB}C_{BA}$ is the weight between the two sides. We prove that X_A is positive and bounded by $\frac{1}{4}$. It is two-sided: X_A and X_B share their positive spectrum. It is stable in operator norm with a certified margin. It lies below the local defect $C_A - C_A^2$, and equals it on a pure state. Four times its trace bounds the Gaussian mutual information from below, with a sharp constant and no upper constant without a spectral margin. The signed boundary offset $\tau = \log\det(I - C_A)/\det C_A$ moves no faster than $\dot{\tau}^2 \le 4gS$. Two preparations with the same census, offset, cross weight and coupling saturate this bound in opposite directions.

On the PT-symmetric Su–Schrieffer–Heeger chain the correlation matrix is a biorthogonal idempotent. There the radial census survives and positivity does not. The band-averaged Petermann factor equals 2 exactly on the line $E(0)E(\pi) = u^2$, equivalently $(w^2-v^2)^2 = 2u^2(w^2+v^2)$.

We construct an operator $T_R = \frac{1}{2}\{X + \frac{1}{2}, H\}_R$ that commutes exactly with the half-chain correlation block. The proof uses only $H(k)^2 = E(k)^2 I$, so it holds without Hermiticity. On its eigenvectors the entanglement levels form a ladder $\varepsilon_j = j\Delta + i\theta$. The spacing is $\Delta = 2\pi K'(k)/K(k)$, with modulus $k = v'/w'$ taken from the Hermitian band hidden in the opened one. The single shared phase is $\theta = \pi F(\arcsin s \,|\, k)/K(k)$, with $s = uw/(w^2 - v'^2)$. We prove that a ladder of this form crosses the unit circle of the Cayley chart all at once, and that its phase reaches $\pi/2$ exactly on the Petermann line. Both directions are proved, the converse through an explicit factorisation. The ladder form and the phase formula are verified at 200 random parameter points: commutation to 8×10^{-12}, spacing to 2×10^{-8}, phase to 4×10^{-7}, and the inside/outside verdict 200 of 200. They are proved in the dimer limit. Deriving them analytically from the commuting operator is the one step left open, and it is stated as such.

Grades. [T] proved here; [L] also checked in Lean in the author's formal suite; [V] verified numerically with the stated controls; [R] a known result in this coordinate; [C] open.

1 Introduction

Every finite record of a cut has two readings. One is unsigned: it counts how many boundary levels are partially occupied. The other is signed: it says which way each level leans. Both readings live

on one chart,

$$\Gamma = \frac{z-1}{z+1} = \tanh\frac{\varepsilon}{2} = 1-2\nu, \qquad z = \frac{1-\nu}{\nu} = e^{\varepsilon},$$

for an occupation ν of the region's correlation matrix. The census is the gauge without its sign. The offset is the clock face. This paper proves two things about this chart.

The first is about the *weight between* the two sides of a cut, the cross block $X_A = C_{AB}C_{BA}$. Section 3 proves its order, size, symmetry, stability and information bounds. It also gives exact examples marking where the one-body record stops speaking, and proves the speed limit of the signed offset.

The second is about where *opened* matter leaves the disk. Opening means a non-Hermitian generator. On the PT-symmetric dimerized chain the levels leave $|\Gamma| < 1$ inside the gapped phase, before the critical point. The full edition of this work located that exit numerically on the line $E(0)E(\pi) = u^2$, where the band-averaged Petermann factor equals 2, and left open why it happens there [1]. Section 5 gives the mechanism. An operator commutes exactly with the half-chain block, and its eigenvectors organise the levels into a ladder with one shared phase. A ladder with one phase crosses the rim all at once, and its phase reaches a quarter turn exactly on the Petermann line. The exact commutation, the simultaneous crossing, the location of the crossing and a closed dimer case are proved. The ladder form and the elliptic phase formula are measured at 200 random points. Deriving them analytically is the remaining step.

2 The chart

Proposition 2.1 (the bridge)**.** $\Gamma = \tanh(\varepsilon/2) = 1-2\nu$*, and* $\nu(1-\nu) = (1-\Gamma^2)/4$*. The window* $\nu \in [\delta, 1-\delta]$ *is the disk* $|\Gamma| \le 1-2\delta$*. [T][L]*

Proof. $z-1 = (1-2\nu)/\nu$ and $z+1 = 1/\nu$; $(e^{\varepsilon}-1)/(e^{\varepsilon}+1) = \tanh(\varepsilon/2)$; $1-(1-2\nu)^2 = 4\nu(1-\nu)$. □

Summed over the levels of the region, the rapidity is the signed offset

$$\tau = \sum_{\nu} \varepsilon_\nu = \mathrm{Tr}[\log(I-C) - \log C] = \log\frac{\det(I-C)}{\det C}, \tag{1}$$

which for a number-conserving Gaussian state is $\log P(N_A=0)/P(N_A=|A|)$. The reduced state is a product over the eigenmodes of C, so $P(0) = \prod(1-\nu)$ and $P(|A|) = \prod \nu$ [T]. The census counts levels with $\nu(1-\nu)$ above a threshold, and is blind to the sign of Γ. For example, $\mathrm{diag}(0.2, 0.8)$ and $\mathrm{diag}(0.2, 0.2)$ have the same defect $C - C^2 = 0.16I$ and offsets 0 and $\log 16$.

Theorem 2.2 (the Cayley domain)**.** *For* $z \neq -1$, $|z+1|^2 - |z-1|^2 = 4\,\mathrm{Re}\, z$*, so* $|\Gamma| < 1 \iff \mathrm{Re}\, z > 0$ *and* $|\Gamma| = 1 \iff \mathrm{Re}\, z = 0$*. For* $z = e^W$*,* $|\tanh(W/2)| < 1 \iff \cos(\mathrm{Im}\, W) > 0$*; on the principal branch this is* $|\mathrm{Im}\, W| < \pi/2$*. [T]*

Proof. Expand the moduli; $\mathrm{Re}\, e^W = e^{\mathrm{Re}\, W}\cos \mathrm{Im}\, W$. □

The real cases are checked in Lean: a positive ratio lands strictly inside, a negative ratio $S/(-s)$ with $S > s > 0$ lands outside at rim distance $2s/(S-s)$, and $z = -1$ is the pole. A complex ratio does not by itself leave the disk: $z = 1 - i\gamma$ has $|\Gamma|^2 = \gamma^2/(4+\gamma^2) < 1$. Leaving needs the real part of the ratio to change sign. In the rapidity this is a quarter turn of the phase.

3 The weight between

Let $C = C^*$ with $0 \le C \le I$ on $\mathcal{H}_A \oplus \mathcal{H}_B$, of dimensions a, b, and write

$$C = \begin{pmatrix} A_C & B_C \\ B_C^* & D_C \end{pmatrix}, \quad F_A = A_C - A_C^2, \quad T_A = (C - C^2)_{AA}, \quad X_A = B_C B_C^*, \quad X_B = B_C^* B_C.$$

Fix $0 < \varepsilon < \frac{1}{2}$ and put $\vartheta = \varepsilon(1-\varepsilon)$. The *raw census* is $N_\varepsilon^{\mathrm{raw}} = \operatorname{rank} \mathbf{1}_{[\varepsilon, 1-\varepsilon]}(A_C)$. The *boundary census* is $N_\varepsilon^{\partial} = \operatorname{rank} \mathbf{1}_{[\vartheta, \infty)}(X_A)$. Counts include multiplicity and equality at the threshold.

Theorem 3.1 (the weight between)**.** *For every such C:*

1. *(defect split) $F_A = T_A + X_A$; with $C^2 = C$, $X_A = A_C - A_C^2$, and this needs no Hermiticity. [T][L]*
2. *(order and size) $0 \le X_A \le F_A \le \frac{1}{4} I$, $0 \le T_A \le F_A$, $\|B_C\| \le \frac{1}{2}$, attained by $C = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. [T][L]*
3. *(two-sided) X_A and X_B have the same positive eigenvalues with multiplicity. So $N_\varepsilon^{\partial}(A) = N_\varepsilon^{\partial}(B) \le \operatorname{rank} B_C \le \min(a, b)$, and $\operatorname{Tr} X_A = \operatorname{Tr} X_B = \|B_C\|_F^2 \le \frac{1}{4}\min(a, b)$. [T]*
4. *(comparison and recovery) $N_\varepsilon^{\partial} \le N_\varepsilon^{\mathrm{raw}}$, with equality when $C^2 = C$. [T]*
5. *(trace) $\vartheta N_\varepsilon^{\partial} \le \operatorname{Tr} X_A = \sum_{i \in A, j \in B} |C_{ij}|^2$, and $\operatorname{Tr} X_A = 0 \iff B_C = 0$. [T][L]*
6. *(local unitaries) Conjugation by $U_A \oplus U_B$ leaves every census, spectrum and trace above unchanged. [T]*

Proof. (1) $(C^2)_{AA} = A_C^2 + B_C B_C^*$.

(2) Compression preserves order. $C - C^2 \ge 0$ by the spectral theorem, hence $T_A \ge 0$. Then (1) gives the two comparisons, and $F_A = \frac{1}{4} I - (A_C - \frac{1}{2})^2 \le \frac{1}{4} I$, so $\|B_C\|^2 = \|X_A\| \le \frac{1}{4}$.

(3) If $B_C B_C^* u = \lambda u$ with $\lambda > 0$, then $B_C^* u \ne 0$ is an eigenvector of $B_C^* B_C$ with eigenvalue λ. The inverse map is B_C/λ, so the multiplicities agree.

(4) $\nu(1-\nu) - \vartheta = (\nu - \varepsilon)(1 - \varepsilon - \nu)$, so $\operatorname{rank} \mathbf{1}_{[\vartheta, \infty)}(F_A) = N_\varepsilon^{\mathrm{raw}}$. By min–max, $\lambda_j(X_A) \le \lambda_j(F_A)$. If $C^2 = C$ then $T_A = 0$.

(5) Each counted eigenvalue contributes at least ϑ to the trace.

(6) Block multiplication. □

The boundary coordinate. Each eigenvalue $\lambda \in [0, \frac{1}{4}]$ of X_A has the radius $\Gamma_b = \sqrt{1 - 4\lambda} \in [0, 1]$. On a pure state $\lambda = \nu(1-\nu)$, so $\Gamma_b = |1 - 2\nu|$: the boundary coordinate is the modulus of the census point and forgets its sign [T][L]. For mixed states X_A need not commute with A_C, and only the order comparisons remain.

Theorem 3.2 (stability with a certified count)**.** *Let $C, \tilde{C}$ be positive contractions on the same partition, with $\delta = \|C - \tilde{C}\|$. Then $\|X_A(C) - X_A(\tilde{C})\| \le \delta$. With $\mathcal{N}_Y(s) = \operatorname{rank} \mathbf{1}_{[s, \infty)}(Y)$, for $\delta < \vartheta$,*

$$\mathcal{N}_{X_A(C)}(\vartheta + \delta) \le N_\varepsilon^{\partial}(\tilde{C}) \le \mathcal{N}_{X_A(C)}(\vartheta - \delta).$$

If $\operatorname{dist}(\vartheta, \operatorname{spec} X_A(C)) > \delta$ the two counts are equal. [T]

Proof. $\|BB^* - \tilde{B}\tilde{B}^*\| \le (\|B\| + \|\tilde{B}\|)\|B - \tilde{B}\| \le \delta$ by Theorem 3.1(2). Then apply Weyl's inequality. □

Theorem 3.3 (near-projector comparison)**.** *With* $\eta = \|C - C^2\|$, $X_A \le F_A \le X_A + \eta I$. *Hence* $\mathcal{N}_{F_A}(\vartheta + \eta) \le N_\varepsilon^\partial \le N_\varepsilon^{\mathrm{raw}}$. *[T]*

Proof. $0 \le C - C^2 \le \eta I$ compresses to $0 \le T_A \le \eta I$. Then use (1) and ordered-eigenvalue monotonicity. □

Proposition 3.4 (the coordinate is Hölder)**.** $|\Gamma_b(\lambda) - \Gamma_b(\tilde{\lambda})| \le 2\sqrt{|\lambda - \tilde{\lambda}|}$. *If both radii are at least* $g > 0$, *the bound is* $(2/g)|\lambda - \tilde{\lambda}|$. *The eigenvalue step does not transfer to nonnormal matrices:* $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ *and* $\begin{pmatrix} 0 & 1 \\ \delta & 0 \end{pmatrix}$ *differ by* δ, *while their eigenvalues move by* $\sqrt{\delta}$. *[T]*

Proof. $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$; rationalise for the second bound. □

What the one-body record cannot say.

Proposition 3.5 (same local block, different weight between)**.** *Every positive contraction* A *has the projector extension* $P_A = \begin{pmatrix} A & \sqrt{A - A^2} \\ \sqrt{A - A^2} & I - A \end{pmatrix}$, *with* $X_A(P_A) = A - A^2$. *The disconnected* $A \oplus (I - A)$ *has the same local block and* $X_A = 0$. *[T]*

Proof. The square root commutes with A. Multiplying the blocks gives $P_A^2 = P_A = P_A^*$. □

Three more exact cases mark the edge of the record [T]:

- $C = pI$ has full raw census and zero boundary census.
- $\frac{1}{2}(|10\rangle\langle 10| + |01\rangle\langle 01|)$ has $C = \frac{1}{2}I_2$ and $X_A = 0$, yet covariance $-\frac{1}{4}$ and mutual information $\log 2$.
- $(|1100\rangle + |0011\rangle)/\sqrt{2}$ split $\{1,2\}|\{3,4\}$ has $X_A = 0$ and entanglement entropy $\log 2$.

The census reads correlation faithfully inside the quasi-free class, and only there.

Proposition 3.6 (number covariance)**.** *For a gauge-invariant quasi-free state,* $\mathrm{Var}(N_A) = \operatorname{Tr} F_A$ *and* $\mathrm{Cov}(N_A, N_B) = -\operatorname{Tr} X_A$. *On a pure Slater state,* $\mathrm{Var}(N_A) = \operatorname{Tr} X_A$. *[T]*

Proof. By Wick's theorem, $\mathrm{Cov}(n_i, n_j) = \delta_{ij} C_{ii} - |C_{ij}|^2$. Sum over $A \times A$ and over $A \times B$. □

Theorem 3.7 (the weight between bounds the information across the cut)**.** *For the same states, with* $h(x) = -x \log x - (1 - x)\log(1 - x)$ *and* $I_G = \operatorname{Tr} h(A_C) + \operatorname{Tr} h(D_C) - \operatorname{Tr} h(C)$*:*

$$4 \operatorname{Tr} X_A \le I_G(A : B),$$

and the constant is sharp. If $\delta I \le C \le (1 - \delta)I$, *then* $I_G \le \operatorname{Tr} X_A/(\delta(1 - \delta))$. *Without such a margin there is no upper constant. [T]*

Proof. Let $S = I_A \oplus (-I_B)$ and $C_0 = \frac{1}{2}(C + SCS) = A_C \oplus D_C$. Trace Jensen states that for concave f, $\operatorname{Tr} f(\frac{1}{2}(H + K)) \ge \frac{1}{2}\operatorname{Tr} f(H) + \frac{1}{2}\operatorname{Tr} f(K)$. To prove it, diagonalise the mean in a basis u_i, then use scalar concavity and $f(\langle u, Hu\rangle) \ge \langle u, f(H)u\rangle$.

Apply it to $f = h + 2x^2$, which has $f'' = 4 - 1/(x(1-x)) \le 0$. With unitary invariance under S, this gives $I_G \ge 2(\operatorname{Tr} C^2 - \operatorname{Tr} C_0^2) = 4 \operatorname{Tr} X_A$. On $[\delta, 1 - \delta]$ the function $h + x^2/(2\delta(1 - \delta))$ is convex, and the same argument gives the upper bound.

Sharpness: $C = \frac{1}{2}I + r\sigma_x$ has $I_G = 4r^2 + O(r^4)$ and $\operatorname{Tr} X_A = r^2$. For the failure without a margin, pure states with an occupation tending to 0 have $I_G/\operatorname{Tr} X_A$ unbounded. □

Theorem 3.8 (the offset speed limit)**.** *Let $P(t)$ be a smooth family of orthogonal projectors and E an isometry onto the region. Set $C = E^\dagger P E$ with $0 < C < I$, $g = \frac{1}{2}\|\dot{P}\|_F^2$ and $S = \mathrm{Tr}[(C(I-C))^{-1}]$. Then $\dot{\tau}^2 \le 4gS$, with equality iff $\dot{P} \propto [P,[P,F]]$, where $F = EWE^\dagger$ and $W = -(C(I-C))^{-1}$. For $\dot{P} = -i[H,P]$, $g = \|PH(1-P)\|_F^2$. [T]*

Proof. $\dot{\tau} = \mathrm{Tr}[W\dot{C}] = \mathrm{Tr}[F\dot{P}]$. Differentiating $P^2 = P$ shows that $\dot{P}$ is block off-diagonal, so only $T = [P,[P,F]]$ pairs with it. Cauchy–Schwarz gives $\dot{\tau}^2 \le \|T\|_F^2\|\dot{P}\|_F^2$. Finally, $\|T\|_F^2 = 2\,\mathrm{Tr}[W(I-C)WC] = 2S$, because W and C commute. For the corollary, $[H,P]$ has two adjoint blocks. □

Proposition 3.9 (the speed budget does not choose the future)**.** *Take $H = \omega\sigma_x$ and $P_\pm(0) = \frac{1}{2}(I \pm \sigma_y)$, read through the first basis vector. Both have $\nu = \frac{1}{2}$, $\tau = 0$, $X = \frac{1}{4}$ and $g = \omega^2$. Then $\tau_\pm(t) = \mp 2\,\mathrm{artanh}(\sin 2\omega t)$: both saturate $\dot{\tau}^2 = 4gS$ at all times, in opposite directions. [T][V]*

Proof. The Bloch vector $(0, \pm 1, 0)$ rotates in the y–z plane at rate 2ω. With $s = \sin 2\omega t$, $S = 4/(1-s^2)$ and $\dot{\tau}^2 = 16\omega^2/(1-s^2) = 4gS$. □

The cross weight of a fixed cut is not the generator coupling of the moving projector. At $t = 0$, $2\sqrt{\mathrm{Tr}\,X \cdot S} = 2$ while $|\dot{\tau}| = 4\omega$. Script `s16` checks the saturation to 5.9×10^{-15}. It also checks every inequality of this section on 400 random contractions and 64 interior covariances. All bounds hold, the sharp ratio runs 4.00027, 4.0000027, 4.00000002, and the pure-state recovery holds to 6.8×10^{-16} [V].

4 Opened matter

Take the PT-symmetric SSH chain, cells $c \in \mathbb{Z}$, two sites a, b per cell:

$$h(k) = \begin{pmatrix} iu & v + we^{-ik} \\ v + we^{ik} & -iu \end{pmatrix}, \qquad E(k)^2 = v^2 + w^2 - u^2 + 2vw\cos k,$$

with $u, v, w > 0$, $v < w$, and the gap condition $E(\pi)^2 = (w-v)^2 - u^2 > 0$. In real space H has $\pm iu$ on the a and b sites, v inside a cell and w from (c,b) to $(c+1,a)$. Then $H^T = H$, but $H^\dagger \ne H$. Fill the lower band with the biorthogonal projector $P(k) = \frac{1}{2}(I - h(k)/E(k))$, $E > 0$, and let C_R be its block on the half-chain $c \ge 0$.

Theorem 4.1 (what survives opening)**.** *For any idempotent $C = \begin{pmatrix} A & B \\ E & D \end{pmatrix}$, $Y_A = BE = A - A^2$, and its eigenvalues are $\nu_j(1-\nu_j)$ with algebraic multiplicity. So the radial census $\#\{|1-2\nu_j| \le r\}$ survives for every r. Positivity does not: $\begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}$ is idempotent with $Y_A = -2$. [T][L]*

Proof. The AA block of $C^2 = C$ gives the first identity. Triangularising A gives the eigenvalues. □

Proposition 4.2 (the Petermann line)**.** *The band projectors have $K(k) = \mathrm{Tr}\,P^\dagger P = 1 + u^2/E(k)^2$. The band average is*

$$\langle K \rangle = 1 + \frac{u^2}{E(0)E(\pi)},$$

so $\langle K \rangle = 2$ iff $E(0)E(\pi) = u^2$, iff

$$(w^2 - v^2)^2 = 2u^2(w^2 + v^2). \tag{2}$$

[T][L]

Proof. $\operatorname{Tr} P^\dagger P = \frac{1}{2}(1 + \sum |d_j|^2/|E|^2)$ for $h = d \cdot \sigma$. Here $\sum |d_j|^2 = |v + we^{ik}|^2 + u^2$ and $|E|^2 = |v+we^{ik}|^2 - u^2$. The average is $\frac{1}{2\pi}\int dk/(A+B\cos k) = (A^2-B^2)^{-1/2}$ with $A^2 - B^2 = E(0)^2E(\pi)^2$. Squaring $E(0)^2E(\pi)^2 = u^4$ gives $((v+w)^2-u^2)((w-v)^2-u^2) = u^4$, that is $(w^2-v^2)^2-2u^2(w^2+v^2) = 0$. □

At $w = 1$, (2) reads $u_{\text{exit}} = (1 - v^2)/\sqrt{2(1 + v^2)}$. At $u = 0.5$ this gives $v_{\text{exit}} = 0.4682$. The full edition measured the census exit of finite blocks on this line to 4×10^{-8} at three values of u. That result needs whole-cell blocks and endpoint filters down to 10^{-9} [V] [1]. Why the census leaves there was the open question, and the next section answers it.

5 The exit ladder

5.1 The Hermitian band inside the opened one

Lemma 5.1. *Put $E_0 = E(0)$, $E_\pi = E(\pi)$, $w' = \frac{1}{2}(E_0 + E_\pi)$, $v' = \frac{1}{2}(E_0 - E_\pi)$. Then $v'^2 + w'^2 = v^2 + w^2 - u^2$, $v'w' = vw$ and*

$$E(k)^2 = |v' + w'e^{ik}|^2,$$

the band of a Hermitian dimerized chain with hoppings $v' < w'$. Its modulus is $k = v'/w'$, and $k' = \sqrt{1-k^2} = \sqrt{E_0E_\pi}/w'$. [T]

Proof. $E_0^2+E_\pi^2 = 2(v^2+w^2-u^2)$ and $E_0^2-E_\pi^2 = 4vw$. Then $v'^2+w'^2 = \frac{1}{2}(E_0^2+E_\pi^2)$, $v'w' = \frac{1}{4}(E_0^2-E_\pi^2)$ and $w'^2 - v'^2 = E_0E_\pi$. □

5.2 An operator that commutes with the half-chain block

Let X be the cell-position operator, $X_{(c,\alpha),(c',\beta)} = c\,\delta_{cc'}\delta_{\alpha\beta}$.

Theorem 5.2 (the commuting operator)**.** *Let $T = \frac{1}{2}\{X+\frac{1}{2}, H\}$. Then $[P, T] = 0$, and $T = T_L \oplus T_R$ has no entries across the cut between cells -1 and 0. Consequently*

$$[C_R, T_R] = 0,$$

entrywise, as an identity of infinite matrices. T_R is tridiagonal, with

$$T_R|_{(c,a),(c,a)} = iu(c+\tfrac{1}{2}), \qquad T_R|_{(c,b),(c,b)} = -iu(c+\tfrac{1}{2}),$$
$$T_R|_{(c,a),(c,b)} = v(c+\tfrac{1}{2}), \qquad T_R|_{(c,b),(c+1,a)} = w(c+1),$$

and it is symmetric. [T]

Proof. The gap condition gives $E(k)^2 \geq E_\pi^2 > 0$, so $P(k)$ is analytic and periodic. The real-space entries $P_{cc'} = \frac{1}{2\pi}\int e^{ik(c-c')}P(k)\,dk$ therefore decay exponentially, and every product below is an absolutely convergent sum. Integration by parts gives $[X, P]_{cc'} = (c-c')P_{cc'} = \frac{i}{2\pi}\int e^{ik(c-c')}\partial_k P(k)\,dk$. Since $[P, H] = 0$,

$$[P, \{X, H\}] = \{[P, X], H\} \iff -i\{\partial_k P, h\} = \tfrac{i}{2}\{\partial_k(h/E), h\}.$$

Differentiating $(h/E)^2 = I$ gives $\{\partial_k(h/E), h/E\} = 0$, so the right side vanishes. The shift by $\frac{1}{2}$ commutes with P trivially, so $[P, T] = 0$.

The entry $T_{ij} = \frac{1}{2}(x_i + x_j + 1)H_{ij}$ crosses the cut only for $i = (-1, b)$, $j = (0, a)$, where $x_i + x_j + 1 = 0$. So $T = T_L \oplus T_R$. Then $(PT)_{RR} = C_RT_R + P_{RL}T_{LR} = C_RT_R$ and $(TP)_{RR} = T_RC_R$. The listed entries follow from those of H. □

The construction is that of Peschel and of Eisler and Peschel for Hermitian chains [2, 3, 5, 6]. What is new here is that it uses only $h(k)^2 = E(k)^2 I$. That relation survives the non-Hermitian opening, so the commuting operator survives with it. Since C_R is bounded and T_R is banded, C_R maps each eigenspace of T_R into itself. A simple eigenvector of T_R is therefore an eigenvector of C_R, and the levels of C_R can be indexed by those of T_R.

5.3 The ladder

Measured statement 5.3 (the edge ladder)**.** Let $K = K(k)$ and $K' = K(k')$ be the complete elliptic integrals of the first kind for the modulus of Lemma 5.1. Put

$$s = \frac{uw}{w^2 - v'^2} \in (0,1), \qquad \theta = \frac{\pi}{K}\, F(\arcsin s \,|\, k) \in (0,\pi), \qquad \Delta = \frac{2\pi K'}{K}.$$

Equivalently, $\mathrm{sn}(\theta K/\pi \,|\, k) = s$. Then the levels of C_R on the eigenvectors of T_R localised at the cut are

$$\varepsilon_j = \log \frac{1-\nu_j}{\nu_j} = j\Delta + i\theta, \qquad j \in \mathbb{Z}.$$

All levels share one phase, and there is no real offset. [V]

Evidence (script `s15`). The test uses 200 random points with $w = 1$, $v \in (0.02, 0.7)$, $u \in (0.02, 0.9)$ and gap $E_\pi \geq 0.2$, of which 16 lie past the exit line. The block has 90 cells, inside a 6000-momentum chain. Results:

- the eigen-residual of C_R on the edge eigenvectors of T_R is at most 8.4×10^{-12};
- the relative spacing error is at most 1.7×10^{-8};
- the phase error is at most 4.2×10^{-7} on the levels $j = 0, \pm 1$;
- the domain condition $s < 1$ holds, with maximum 0.913 at the test points and $1 - 4 \times 10^{-8}$ on a 400×400 grid of the whole gapped region, where $s \to 1$ only at the critical line.

Deeper levels have ν within 10^{-10} of 0 or 1. There the nonnormal eigenproblem is ill-conditioned, and they are not used. At $u = 0$, $k = v/w$ and the spacing is the elliptic spacing of the Hermitian dimerized chain known from corner-transfer-matrix methods [3, 4], with the protected $\nu = \frac{1}{2}$ edge level at $j = 0$ [R]. Opening the chain keeps the spacing of the hidden Hermitian band and rotates the whole ladder by one phase.

5.4 The exit theorem

Theorem 5.4 (the exit ladder)**.** *(a) (one phase, one crossing) If $\varepsilon_j = j\Delta + i\theta$ with $\theta \in (0,\pi)$, then every $\Gamma_j = \tanh(\varepsilon_j/2)$ lies on the circular arc through -1, $i\tan(\theta/2)$ and 1. Either all $|\Gamma_j| < 1$, or all $|\Gamma_j| = 1$, or all $|\Gamma_j| > 1$, according as $\theta < \pi/2$, $\theta = \pi/2$ or $\theta > \pi/2$.*

(b) (the crossing is the Petermann line) For θ as in Statement 5.3,

$$\theta = \frac{\pi}{2} \iff E(0)E(\pi) = u^2 \iff \langle K \rangle = 2 \iff (w^2 - v^2)^2 = 2u^2(w^2 + v^2).$$

Moreover $\theta < \pi/2$ on the side $E(0)E(\pi) > u^2$ and $\theta > \pi/2$ on the other.

(c) (the dimer limit, exact) At $v = 0$ the half-chain block has exactly one level off $\{0, 1\}$, with $\varepsilon = 2i \arcsin(u/w)$. This equals the formula of Statement 5.3 at $v = 0$. That level leaves the disk exactly at $w^2 = 2u^2$, which is (2) at $v = 0$.

[T]

Proof. (a) By Theorem 2.2, $|\tanh(W/2)| < 1 \iff \cos \operatorname{Im} W > 0$. Here $\operatorname{Im} \varepsilon_j = \theta$ for every j, in the principal range. The image of the line $\{x + i\theta\}$ under $W \mapsto \tanh(W/2)$ is a circle arc. Its endpoints are ± 1 as $x \to \pm\infty$, and $x = 0$ gives $\tanh(i\theta/2) = i \tan(\theta/2)$.

(b) On $(0, K)$, $\operatorname{sn}(\cdot \,|\, k)$ increases strictly from 0 to 1, and $\operatorname{sn}(K/2 \,|\, k)^2 = 1/(1 + k')$ [7]. So $\theta = \pi/2$ iff $s^2(1 + k') = 1$. Put $W = w'$ and $r = \sqrt{E_0 E_\pi}$, so that $k' = r/W$ (Lemma 5.1). Then

$$\mathcal{F} := u^2 w^2 (W + r) - W(w^2 - v'^2)^2 = 0.$$

Lemma 5.1 makes w^2 and v^2 the roots of $x^2 - (2W^2 - r^2 + u^2)x + W^2(W^2 - r^2) = 0$. With $\delta = u^2 - r^2$ and $D = \sqrt{4W^2u^2 + \delta^2}$, the larger root ($w > v$) is $w^2 = W^2 + \frac{1}{2}(\delta + D)$, and $w^2 - v'^2 = \frac{1}{2}(u^2 + r^2 + D)$. So $\mathcal{F} = \mathcal{F}(D)$ is a polynomial in D. Multiplying by its conjugate $\mathcal{F}(-D)$ and reducing with $D^2 = 4W^2u^2 + \delta^2$ gives the identity

$$\mathcal{F}(D)\, \mathcal{F}(-D) = W\, r^2 u^2\, (W + r)^2\, (W - r)\, (u^2 - r^2).$$

Since $0 < r < W$ and $u > 0$, $\mathcal{F}(D) = 0$ forces $u^2 = r^2 = E(0)E(\pi)$.

Conversely, on $u^2 = r^2$ we have $\delta = 0$, $D = 2Wu$, $w^2 = W(W + u)$ and $w^2 - v'^2 = u(W + u)$. Then $s^2 = w^2/(W+u)^2 = W/(W+u) = 1/(1+k')$. The remaining equivalences are Proposition 4.2.

For the sides: θ is continuous in (v, u) on the gapped region and equals $\pi/2$ only on the line. Each side of the line is connected, since for fixed v it is an interval in u with continuous endpoints. As $u \to 0$, $s \to 0$ and $\theta \to 0$; at $v = 0$, $\theta = 2 \arcsin(u/w) > \pi/2$ for $u^2 > w^2/2$.

(c) At $v = 0$, H is a direct sum of dimers on $\{(c, b), (c + 1, a)\}$ with matrix $\begin{pmatrix} -iu & w \\ w & iu \end{pmatrix}$ and $E = \sqrt{w^2 - u^2}$. The half-chain $c \geq 0$ contains whole dimers, whose blocks are idempotent with $\nu \in \{0, 1\}$, and the single site $(0, a)$ of the cut dimer. For that site, $\nu = \frac{1}{2}(1 - iu/E)$, so $z = (E + iu)/(E - iu)$ has $|z| = 1$ and $\varepsilon = 2i \arctan(u/E) = 2i \arcsin(u/w)$. At $v = 0$ we have $v' = 0$, $k = 0$, $K = \pi/2$ and $F(\phi \,|\, 0) = \phi$, so $s = u/w$ and $\theta = 2 \arcsin(u/w)$. The level leaves where $2 \arcsin(u/w) = \pi/2$, that is $w^2 = 2u^2$, and $E(0)E(\pi) = w^2 - u^2 = u^2$ there. □

Sympy confirms that $\mathcal{F}$ vanishes on the line, and that the product reduces to the factored form above. Script `s15` returns $|\theta - \pi/2| \leq 6.7 \times 10^{-16}$ on the line at six values of v, and the dimer limit to 2.2×10^{-16} [V].

5.5 What the theorem explains, and its scope

1. *Why the exit is filter-independent on this chain.* The levels nearest the endpoints, $\operatorname{Re} \varepsilon_j \to \pm\infty$, cross with the centre level. A filter that discards them does not move the crossing until it begins to discard ill-conditioned levels it should keep. This matches the full edition's filter table, where the massless exit holds through 10^{-9} [1].

2. *Why the quarter-turn continuation fits only at the exit.* The continued endpoint offset $\Gamma_{\text{end}} = i\gamma$, $\gamma = u/\sqrt{E_0 E_\pi}$, is the apex $i \tan(\theta/2)$ of the arc only at $v = 0$ and on the exit line. Elsewhere $\tan(\theta/2) \neq \gamma$ (Figure 1, dashed). This is the measured excess of the rotation over the continuation that the full edition reported.

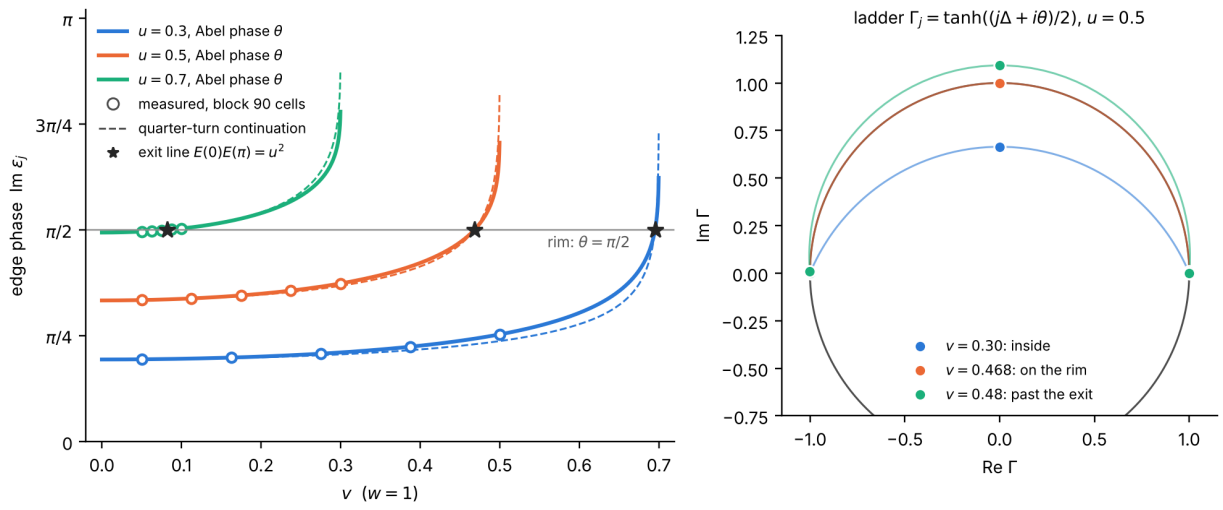


Figure 1: Left: the shared edge phase θ from Statement 5.3 (solid) against v at three values of u ($w = 1$). Open circles: measured $\operatorname{Im}\varepsilon_0$, block of 90 cells. Dashed: the quarter-turn continuation $2\arctan(u/\sqrt{E_0 E_\pi})$. Stars: the exit line (2), where every solid curve crosses $\pi/2$. Right: the ladder Γ_j, $j = -3, \ldots, 3$, at $u = 0.5$, inside, on and past the exit. Each ladder sits on one arc through ± 1, so it crosses the rim all at once.

3. *Partition.* The cut runs between cells. With $w = 0$ every whole-cell block is idempotent and nothing leaves, even where $\langle K \rangle = 2$. That is consistent with (c): the level that exits belongs to a dimer cut in two.

4. *Other bands.* Theorem 5.2 uses only $h(k)^2 = E(k)^2 I$ and nearest-cell hopping, so the commuting operator exists for any such two-band chain. Statement 5.3 and part (b) are stated for real u only, where the on-site term is purely imaginary. The full edition's massive band, $d = (t_1 + \cos k, \sin k, m + i\gamma)$, has a complex on-site term. Its exit is earlier and filter-dependent [1]. Whether its ladder still carries a single phase is not tested here.

The open step. Theorem 5.2 fixes the eigenvectors. Theorem 5.4 turns any ladder of the measured form into the exit law. What remains is to derive the ladder form and the elliptic phase of Statement 5.3 from T_R analytically. In the Hermitian case this is done by the corner transfer matrix. Here the natural route is the Wiener–Hopf factorisation of the band symbol on the elliptic curve $y^2 = -z(v' + w'z)(v'z + w')$. On that curve the phase appears as the Abel image of the point $z = -v/w$, where the opened eigenvector $(E - iu, -(v + wz))$ degenerates. That derivation is open [C]. With it, the exit law on the coupled PT-symmetric chain would be a theorem end to end.

6 Status

Result	Grade
Bridge	[T][L]
Cayley domain (real cases also in Lean)	[T]
Weight between: split, order, two-sidedness, comparison, trace, local unitaries	[T][L]
Operator-norm stability, near-projector comparison, Hölder radius	[T]
Projector extension and three exact limits of the one-body record	[T]
$4 \operatorname{Tr} X_A \le I_G$, sharp; upper bound with margin, none without	[T][V]
Offset speed limit and the opposite-direction saturating pair	[T][V]
Radial census survives opening; positivity does not	[T][L]
Petermann line $\langle K \rangle = 2 \iff (w^2 - v^2)^2 = 2u^2(w^2 + v^2)$	[T][L]
Hidden Hermitian band; commuting operator $[C_R, T_R] = 0$	[T]
Ladder crosses the rim all at once; its phase is $\pi/2$ exactly on the Petermann line (both directions); dimer limit	[T]
Ladder form $\varepsilon_j = j\Delta + i\theta$, elliptic spacing and Abel phase, 200 random points	[V]
Finite-block census exit on the line, to 4×10^{-8} (full edition)	[V]
Analytic derivation of the ladder from T_R	[C]

References

[1] J. Beasley, *Matter, Then On Down: one coordinate from bulk matter to the nucleus*, Edition 4 (2026), §§2, 7 and Appendix C.

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[3] I. Peschel, On the reduced density matrix for a chain of free electrons, *J. Stat. Mech.* (2004) P06004.

[4] I. Peschel and V. Eisler, Reduced density matrices and entanglement entropy in free lattice models, *J. Phys. A* **42** (2009) 504003.

[5] V. Eisler and I. Peschel, Free-fermion entanglement and spheroidal functions, *J. Stat. Mech.* (2013) P04028.

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[7] P. F. Byrd and M. D. Friedman, *Handbook of Elliptic Integrals for Engineers and Scientists*, 2nd ed., Springer (1971).

A Conventions

Logarithms are natural. $K(k) = F(\pi/2 \,|\, k)$ with $F(\phi \,|\, k) = \int_0^\phi (1 - k^2 \sin^2 t)^{-1/2} dt$ (the scripts call `mpmath` with parameter $m = k^2$). The biorthogonal projector uses the principal square root $E(k) > 0$, which is valid throughout the gapped region. The cut lies between cells -1 and 0, and the region is $c \ge 0$; in the scripts it is the left edge of a finite block of 90 cells. The phases $\operatorname{Im} \varepsilon$ use the principal logarithm.

B Scripts

Each listing prints the numbers quoted in the text. The outputs are reproduced after each listing.

B.1 s15_exit_ladder.py

```
# SCRIPT: s15_exit_ladder
"""s15: the exit ladder on the opened PT-symmetric SSH chain.
Checks, at 200 random gapped parameter points (w=1, seed 15):
  (a) T_R = 1/2{X+1/2,H} restricted to the right half commutes with the biorthogonal block C_R
      (measured as the eigen-residual of C_R on edge-localised eigenvectors of T_R);
  (b) the edge levels are eps_j = j*Delta + i*theta with Delta = 2*pi*K'(k)/K(k), k = v'/w';
  (c) theta = pi*F(arcsin s | k^2)/K(k), s = u*w/(w^2 - v'^2);
  (d) the ladder is inside the disk iff E(0)E(pi) > u^2 (sign of cos theta).
Also the exact dimer case v=0, theta = 2 arcsin(u/w)."""
import numpy as np, mpmath as mp
rng=np.random.default_rng(15)
def block(v,w,u,Lc,N=6000):
    ks=2*np.pi*(np.arange(N)+0.5)/N
    E=np.sqrt(v*v+w*w-u*u+2*v*w*np.cos(ks)+0j)
    q=v+w*np.exp(-1j*ks) # h = [[iu, q],[conj-in-k q~, -iu]]
    qt=v+w*np.exp(1j*ks)
    P=np.empty((N,2,2),complex) # lower band P = (I - h/E)/2
    P[:,0,0]=0.5*(1-1j*u/E); P[:,1,1]=0.5*(1+1j*u/E)
    P[:,0,1]=-0.5*q/E; P[:,1,0]=-0.5*qt/E
    d=np.arange(-(Lc-1),Lc)
    G=np.einsum('dk,kab->dab',np.exp(1j*np.outer(d,ks)),P)/N
    C=np.zeros((2*Lc,2*Lc),complex)
    for c in range(Lc):
        for cp in range(Lc):
            C[2*c:2*c+2,2*cp:2*cp+2]=G[c-cp+Lc-1]
    return C
def Tmat(v,w,u,L):
    T=np.zeros((2*L,2*L),complex)
    for c in range(L):
        a,b=2*c,2*c+1; x=c+0.5
        T[a,a]=1j*u*x; T[b,b]=-1j*u*x; T[a,b]=T[b,a]=v*x
        if c<L-1: T[b,a+2]=T[a+2,b]=w*(c+1)
    return T
def predict(v,w,u):
    E0=np.sqrt((v+w)**2-u*u); Ep=np.sqrt((w-v)**2-u*u)
    wp=(E0+Ep)/2; vp=(E0-Ep)/2; k=vp/wp
    K=float(mp.ellipk(k*k)); Kp=float(mp.ellipk(1-k*k))
    s=u*w/(w*w-vp*vp)
    th=np.pi*float(mp.ellipf(mp.asin(s),k*k))/K
    return 2*np.pi*Kp/K, th, E0*Ep
def measure(v,w,u,L=90):
    C=block(v,w,u,L); T=Tmat(v,w,u,L)
    ev,R=np.linalg.eig(T); lv=[]
    for i in np.argsort(abs(ev))[:12]:
        r=R[:,i]/np.linalg.norm(R[:,i])
        if np.abs(r[-30:]).max()>1e-9: continue
        nu=(r.conj()@C@r)
        if abs(nu)<1e-14 or abs(1-nu)<1e-14: continue
        res=np.linalg.norm(C@r-nu*r)
        lv.append((ev[i].real,np.log((1-nu)/nu),res))
    lv.sort(key=lambda x:x[0])
    return lv
out=[]
pts=[]
while len(pts)<200:
    v=rng.uniform(0.02,0.7); u=rng.uniform(0.02,0.9)
    Ep2=(1-v)**2-u*u
    if Ep2<0.2**2 : continue
    pts.append((v,u))
worst={'res':0,'phase':0,'spacing':0}; agree=0; outside=0
```

```
for (v,u) in pts:
    D,th,p=predict(v,1.0,u)
    lv=measure(v,1.0,u)
    if not lv:
        print('no␣edge␣levels␣at',v,u); continue
    j0=min(range(len(lv)),key=lambda i:abs(lv[i][1].real))
    use=[x for x in lv if abs(x[1].real)<1.5*D+1e-9 and x[2]<1e-6] # j=0,+-1
    for t,e,res in use:
        worst['res']=max(worst['res'],res)
        worst['phase']=max(worst['phase'],abs(np.angle(np.exp(1j*(e.imag-th)))))
    re=sorted(x[1].real for x in use)
    for a,b in zip(re,re[1:]):
        if D<40: worst['spacing']=max(worst['spacing'],abs((b-a)-D)/D)
    meas_in=np.cos(lv[j0][1].imag)>0; pred_in=p>u*u
    agree+= (meas_in==pred_in); outside+= (not pred_in)
lines=[f'points:␣200␣(w=1,␣seed␣15,␣gap␣E(pi)>=0.2,␣block␣90␣cells),␣of␣which␣{outside}␣lie␣past␣the␣exit␣line
    ␣E0*Epi<u^2',
 f'(a)␣max␣eigen-residual␣of␣C_R␣on␣T_R␣edge␣eigenvectors␣(levels␣j=0,+-1):␣{worst["res"]:.2e}',
 f'(b)␣max␣relative␣spacing␣error␣|Re(eps_(j+1)-eps_j)-Delta|/Delta:␣␣␣␣␣␣␣␣␣{worst["spacing"]:.2e}',
 f'(c)␣max␣phase␣error␣|Im␣eps_j␣-␣theta_pred|␣(j=0,+-1):␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣{worst["phase"]:.2e}',
 f'(d)␣inside/outside␣verdict␣agrees␣with␣sign(E0*Epi-u^2):␣{agree}/200']
# exact dimer case v=0
md=0
for u in [0.1,0.3,0.5,0.6,0.7,0.75,0.9]:
    D,th,p=predict(0.0,1.0,u)
    md=max(md,abs(th-2*np.arcsin(u)))
lines.append(f'v=0␣dimer␣limit:␣max␣|theta_pred␣-␣2␣arcsin(u/w)|␣over␣7␣values␣of␣u:␣{md:.2e}')
# exit line: theta_pred at E0*Epi=u^2 equals pi/2
mx=0
for v in [0.05,0.1,0.2,0.3,0.4,0.5]:
    uu=(1-v*v)/np.sqrt(2*(1+v*v)) # closed form of E0*Epi=u^2 at w=1
    mx=max(mx,abs(predict(v,1.0,uu)[1]-np.pi/2))
lines.append(f'exit␣line:␣max␣|theta_pred␣-␣pi/2|␣on␣E0*Epi=u^2␣at␣6␣values␣of␣v:␣{mx:.2e}')
smax=0
for (v,u) in pts:
    E0=np.sqrt((1+v)**2-u*u); Ep=np.sqrt((1-v)**2-u*u); vp=(E0-Ep)/2
    smax=max(smax,u/(1-vp*vp))
lines.append(f'domain:␣max␣s␣=␣u*w/(w^2-v\'^2)␣over␣the␣200␣points:␣{smax:.4f}␣(<␣1)')
print('\n'.join(lines))
open('../outputs/s15_exit_ladder.txt','w').write('\n'.join(lines)+'\n')
```

```
points: 200 (w=1, seed 15, gap E(pi)>=0.2, block 90 cells), of which 16 lie past the exit line E0*Epi<u^2
(a) max eigen-residual of C_R on T_R edge eigenvectors (levels j=0,+-1): 8.37e-12
(b) max relative spacing error |Re(eps_(j+1)-eps_j)-Delta|/Delta: 1.67e-08
(c) max phase error |Im eps_j - theta_pred| (j=0,+-1): 4.21e-07
(d) inside/outside verdict agrees with sign(E0*Epi-u^2): 200/200
v=0 dimer limit: max |theta_pred - 2 arcsin(u/w)| over 7 values of u: 2.22e-16
exit line: max |theta_pred - pi/2| on E0*Epi=u^2 at 6 values of v: 6.66e-16
domain: max s = u*w/(w^2-v'^2) over the 200 points: 0.9131 (< 1)
```

B.2 s16_weight_between_checks.py

```
# SCRIPT: s16_weight_between_checks
"""s16: numerical controls for the weight-between theorems (journal edition).
Random positive contractions C on a=4, b=5 (seed 16)."""
import numpy as np
rng=np.random.default_rng(16); a,b=4,5; n=a+b
def rand_contraction(n,pure=False):
    Z=rng.normal(size=(n,n))+1j*rng.normal(size=(n,n)); U,_=np.linalg.qr(Z)
    lam=(rng.integers(0,2,n).astype(float) if pure else rng.uniform(0,1,n))
    return U@np.diag(lam)@U.conj().T
h=lambda x: -(x*np.log(x)+(1-x)*np.log(1-x))
def trh(M): e=np.linalg.eigvalsh(M); e=np.clip(e,1e-300,1-1e-16); return np.sum(h(e))
L=[]; w={}
def upd(k,x): w[k]=max(w.get(k,-np.inf),x)
```

```
for _ in range(400):
    C=rand_contraction(n); A=C[:a,:a]; B=C[:a,a:]; D=C[a:,a:]
    X=B@B.conj().T; XB=B.conj().T@B; F=A-A@A
    TA=(C-C@C)[:a,:a]
    upd('defect split |F-T-X|',np.abs(F-TA-X).max())
    ex=np.linalg.eigvalsh(X); ef=np.linalg.eigvalsh(F)
    upd('order: max eig X - 1/4',ex.max()-0.25); upd('order: min eig(F-X)',-np.linalg.eigvalsh(F-X).min())
    pos=lambda e: np.sort(e[e>1e-12])
    upd('two-sided |spec X_A - spec X_B|',np.abs(pos(ex)-pos(np.linalg.eigvalsh(XB))).max())
    # min-max comparison
    upd('ordered lambda_j(X)-lambda_j(F)',(np.sort(ex)-np.sort(ef)).max())
    # stability
    E=rng.normal(size=(n,n))+1j*rng.normal(size=(n,n)); E=(E+E.conj().T)/2; E*=1e-3/np.linalg.norm(E,2)
    Ct=C+E; ev,V=np.linalg.eigh(Ct); Ct=V@np.diag(np.clip(ev,0,1))@V.conj().T
    dl=np.linalg.norm(C-Ct,2); Xt=Ct[:a,a:]@Ct[:a,a:].conj().T
    upd('stability ||X-X~||/delta - 1',np.linalg.norm(X-Xt,2)/dl-1)
# information bound on 64 interior covariances
lo=np.inf; up=np.inf
for _ in range(64):
    C=rand_contraction(n); ev,V=np.linalg.eigh(C); dlt=0.05; ev=dlt+(1-2*dlt)*ev; C=V@np.diag(ev)@V.conj().T
    A=C[:a,:a]; D=C[a:,a:]; X=C[:a,a:]@C[:a,a:].conj().T
    I=trh(A)+trh(D)-trh(C); tx=np.trace(X).real
    lo=min(lo,I-4*tx); up=min(up,tx/(dlt*(1-dlt))-I)
L.append(f'info bound, 64 covariances with spectrum in [0.05,0.95]: min(I_G-4TrX)={lo:.3e}, min(TrX/(d(1-d))-
    I_G)={up:.3e}')
for r in [1e-2,1e-3,1e-4]:
    I=2*np.log(2)-2*h(0.5+r); L.append(f'sharp family C=1/2 I+r sigma_x, r={r:g}: I_G/TrX={I/r**2:.9f}')
# pure states: census recovery
mx=0
for _ in range(100):
    C=rand_contraction(n,pure=True); A=C[:a,:a]; X=C[:a,a:]@C[:a,a:].conj().T
    mx=max(mx,np.abs(X-(A-A@A)).max())
L.append(f'pure states (100): max |X_A-(A_C-A_C^2)| = {mx:.2e}')
# speed-limit saturating pair
sx=np.array([[0,1],[1,0]],complex); sy=np.array([[0,-1j],[1j,0]])
worst=0
for om in [0.5,1.0,2.0]:
    H=om*sx
    for sgn in [+1,-1]:
        P0=0.5*(np.eye(2)+sgn*sy)
        for t in np.linspace(0.01,0.7/om,31):
            if abs(np.sin(2*om*t))>0.95: continue
            U=np.cos(om*t)*np.eye(2)-1j*np.sin(om*t)*sx; P=U@P0@U.conj().T
            Pd=-1j*(H@P-P@H); nu=P[0,0].real; nud=Pd[0,0].real
            taud=-nud/(nu*(1-nu)); g=0.5*np.linalg.norm(Pd)**2; S=1/(nu*(1-nu))
            worst=max(worst,abs(taud**2/(4*g*S)-1))
L.append(f'offset speed limit, pair P+-: max |taudot^2/(4gS)-1| over 3 omegas, 2 signs, 31 times = {worst:.1e}
    ')
for k,x in w.items(): L.insert(0,f'400 random contractions (a=4,b=5): max[{k}] = {x:.2e}')
L.insert(0,'(for order and stability lines, a negative maximum means the bound holds with that margin)')
print('\n'.join(L)); open('../outputs/s16_weight_between_checks.txt','w').write('\n'.join(L)+'\n')
```

```
(for order and stability lines, a negative maximum means the bound holds with that margin)
400 random contractions (a=4,b=5): max[stability ||X-X~||/delta - 1] = -5.66e-01
400 random contractions (a=4,b=5): max[ordered lambda_j(X)-lambda_j(F)] = -2.54e-02
400 random contractions (a=4,b=5): max[two-sided |spec X_A - spec X_B|] = 1.39e-16
400 random contractions (a=4,b=5): max[order: min eig(F-X)] = -2.26e-02
400 random contractions (a=4,b=5): max[order: max eig X - 1/4] = -5.72e-02
400 random contractions (a=4,b=5): max[defect split |F-T-X|] = 1.87e-16
info bound, 64 covariances with spectrum in [0.05,0.95]: min(I_G-4TrX)=2.106e-03, min(TrX/(d(1-d))-I_G)=3.318e
    -01
sharp family C=1/2 I+r sigma_x, r=0.01: I_G/TrX=4.000266709
sharp family C=1/2 I+r sigma_x, r=0.001: I_G/TrX=4.000002667
sharp family C=1/2 I+r sigma_x, r=0.0001: I_G/TrX=4.000000020
pure states (100): max |X_A-(A_C-A_C^2)| = 6.80e-16
offset speed limit, pair P+-: max |taudot^2/(4gS)-1| over 3 omegas, 2 signs, 31 times = 5.9e-15
```

B.3 fig09_exit_ladder.py

```
# SCRIPT: fig09_exit_ladder
"""fig09: the exit ladder. Left: theta(v) from the Abel formula, measured edge phases, quarter-turn
    continuation.
Right: the ladder Gamma_j = tanh((j*Delta + i*theta)/2) in the disk, inside / on / past the exit line."""
import numpy as np, mpmath as mp, matplotlib
matplotlib.use('Agg'); import matplotlib.pyplot as plt
exec(open('s15_exit_ladder.py').read().split('out=[]')[0])
plt.rcParams.update({'font.size':9,'axes.spines.top':False,'axes.spines.right':False,'pdf.fonttype':42})
cols=['#2a78d6','#eb6834','#1baf7a']
fig,ax=plt.subplots(1,2,figsize=(9.2,3.9),gridspec_kw={'width_ratios':[1.35,1]})
a=ax[0]
for u,c in zip([0.3,0.5,0.7],cols):
    vmax=1-u-1e-4; vs=np.concatenate([np.linspace(0.0,vmax-0.05,150),vmax-np.geomspace(0.05,1e-4,150)])
    th=[predict(v,1,u)[1] for v in vs]
    gam=[2*np.arctan(u/np.sqrt(np.sqrt((1+v)**2-u*u)*np.sqrt((1-v)**2-u*u))) for v in vs]
    a.plot(vs,th,color=c,lw=2,label=f'$u={u}$, Abel phase $\\theta$')
    a.plot(vs,gam,color=c,lw=1,ls='--')
    vm=np.linspace(0.05,1-u-0.2,5)
    for v in vm:
        lv=measure(v,1,u); e=min(lv,key=lambda x:abs(x[1].real))[1]
        a.plot(v,e.imag,'o',ms=5,mfc='white',mec=c,mew=1.5)
    ve=float(mp.findroot(lambda x:(1-x*x)/mp.sqrt(2*(1+x*x))-u,0.4))
    a.plot(ve,np.pi/2,'*',color='0.15',ms=9,zorder=5)
a.axhline(np.pi/2,color='0.55',lw=0.8); a.text(0.56,np.pi/2-0.13,'rim: $\\theta=\\pi/2$',color='0.35',fontsize
    =8)
a.set_xlabel('$v$ ($w=1$)'); a.set_ylabel('edge phase  Im $\\varepsilon_j$')
a.set_yticks([0,np.pi/4,np.pi/2,3*np.pi/4,np.pi]); a.set_yticklabels(['0','$\\pi/4$','$\\pi/2$','$3\\pi/4$','$
    \\pi$'])
a.plot([],[],'o',mfc='white',mec='0.3',label='measured, block 90 cells'); a.plot([],[],'--',color='0.3',lw=1,
    label='quarter-turn continuation')
a.plot([],[],'*',color='0.15',label='exit line $E(0)E(\\pi)=u^2$')
a.legend(frameon=False,fontsize=7.5,loc='upper left'); a.set_ylim(0,np.pi*1.02); a.set_xlim(-0.02,0.72)
b=ax[1]; t=np.linspace(0,2*np.pi,400); b.plot(np.cos(t),np.sin(t),color='0.3',lw=1)
u=0.5
for v,c,lab in [(0.3,cols[0],'$v=0.30$: inside'),(0.4682,cols[1],'$v=0.468$: on the rim'),(0.48,cols[2],'$v
    =0.48$: past the exit')]:
   D,th,p=predict(v,1,u); x=np.linspace(-12,12,400)
   G=np.tanh((x+1j*th)/2); b.plot(G.real,G.imag,color=c,lw=1,alpha=0.6)
   j=np.arange(-3,4); Gj=np.tanh((j*D+1j*th)/2)
   b.plot(Gj.real,Gj.imag,'o',color=c,ms=6,mec='white',mew=1,label=lab)
b.set_aspect('equal'); b.set_xlim(-1.15,1.15); b.set_ylim(-0.75,1.25)
b.set_xlabel('Re $\\Gamma$'); b.set_ylabel('Im $\\Gamma$'); b.legend(frameon=False,fontsize=7.5,loc='lower
    center',ncol=1)
b.set_title('ladder $\\Gamma_j=\\tanh((j\\Delta+i\\theta)/2)$, $u=0.5$',fontsize=9)
plt.tight_layout(); plt.savefig('../figures/fig09_exit_ladder.png',dpi=220); plt.savefig('../figures/
    fig09_exit_ladder.pdf')
```

B.4 Symbolic check of Theorem 5.4(b)

```
# SCRIPT: s17_exit_converse_sympy
"""s17: symbolic check of Theorem 5.4(b). W=w', r=sqrt(E0*Epi), D=sqrt(4W^2u^2+(u^2-r^2)^2).
F = u^2 w^2 (W+r) - W (w^2-v'^2)^2 vanishes iff theta = pi/2."""
import sympy as sp
W,r,u,D=sp.symbols('W r u D',positive=True)
d=u**2-r**2
w2=W**2+(d+D)/2; vp2=W**2-r**2
F=sp.expand(u**2*w2*(W+r)-W*(w2-vp2)**2)
prod=sp.expand(sp.expand(F*F.subs(D,-D)).subs(D**2,4*W**2*u**2+d**2))
fac=sp.factor(prod)
target=W*r**2*u**2*(W+r)**2*(W-r)*(u**2-r**2)
lines=[f'F(D)F(-D) factors as: {fac}',
       f'F(D)F(-D) - W r^2 u^2 (W+r)^2 (W-r)(u^2-r^2) = {sp.simplify(prod-sp.expand(target))}',
       f'F on the line u=r (D=2Wu): {sp.simplify(F.subs(D,2*W*u).subs(r,u))}']
```

```
# root-check: w^2 is the larger root of x^2-(2W^2-r^2+u^2)x+W^2(W^2-r^2)
x=sp.symbols('x'); q=x**2-(2*W**2-r**2+u**2)*x+W**2*(W**2-r**2)
lines.append(f'w^2 satisfies the quadratic: {sp.simplify(sp.expand(q.subs(x,w2)).subs(D**2,4*W**2*u**2+d**2))}
    ')
print('\n'.join(lines)); open('../outputs/s17_exit_converse_sympy.txt','w').write('\n'.join(lines)+'\n')
```

```
F(D)F(-D) factors as: W*r**2*u**2*(-W + r)*(W + r)**2*(r - u)*(r + u)
F(D)F(-D) - W r^2 u^2 (W+r)^2 (W-r)(u^2-r^2) = 0
F on the line u=r (D=2Wu): 0
w^2 satisfies the quadratic: 0
```